

SD-WAN Parallel Convergence Test Report

Branches BR8 and BR5 to Data Center DC1 - network probes and custom TCP applications

Test date	Tuesday 22 September 2026, 22:10 to 22:16 UTC
Test IDs	BR8: G-20260923-8EE9 (CONV-0235) BR5: G-20260923-B5AB (CONV-0022)
Sources / Target	BR8-Ubuntu (192.168.219.1) and BR5-Ubuntu (192.168.217.5) to DC1 (192.168.203.100)
Platform	Prisma SD-WAN fabric, VyOS underlay emulation, Stigix traffic & telemetry
Result	PASS with observations - DC1 stayed reachable from both branches through all failures; ≈ 8 s outage per branch at final failover; TCP sessions to DC1 were reset and re-established

1. Executive summary

$\approx 2\text{ s}$

SD-WAN path decision
(BR8, exact timestamp)

$8.7\text{ s} / 8.1\text{ s}$

Equivalent outage
BR8 / BR5 (UDP 6200)

$+2\text{ per app}$

TCP sessions to DC1 reset
on every active app

$< 2\text{ min}$

Automatic failback
both sites

Two branches, BR8 and BR5, were tested in parallel against the same sequence of faults on DC1's WAN transports. Each ran a continuous UDP probe to DC1 (port 6200, 50 packets/s), while the custom TCP applications already deployed on the Stigix nodes provided an application-level view of the impact.

Key findings. Both branches behaved consistently. A +50 ms latency on DC1's Internet overlay raised latency visibly but triggered no path change and no loss, on either the network probes or the TCP applications. Losing DC1 MPLS had no effect. Losing DC1 Internet caused both branches to reroute to DC1 via the DC2 hub and the DC1-DC2 interconnect: BR8's path decision is timestamped about two seconds after the fault, and the probes measured roughly 8 seconds of interruption per branch. Every active TCP application lost its sessions to DC1 and reconnected automatically over the backup path. After restoration, both branches returned to DC1 directly in under two minutes, BR8 on its second Internet circuit rather than its original one.

2. Scope and topology

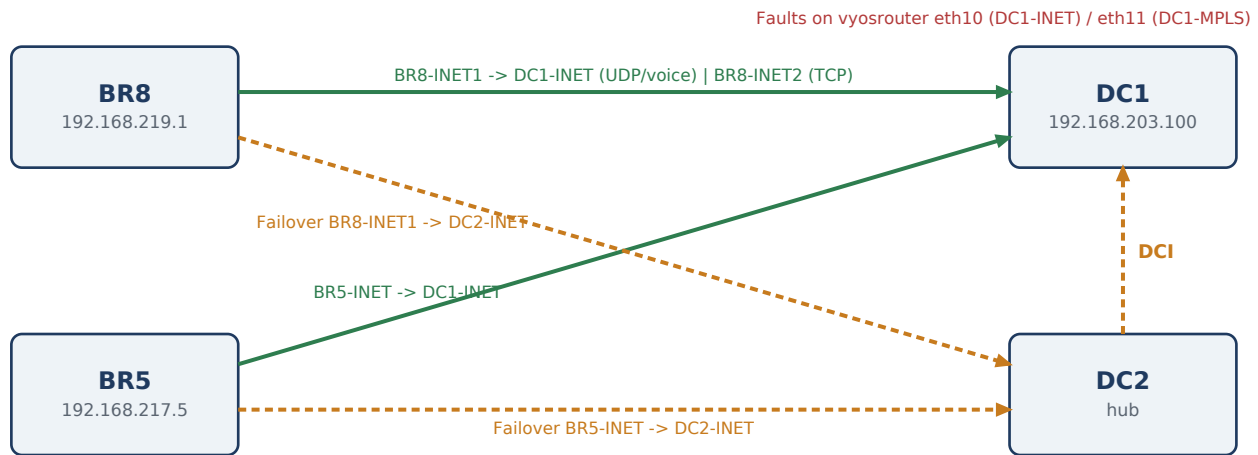


Figure 1 - Nominal paths (green) and failover paths via DC2 and the DCI (orange).

Element	Identifier	Role
Branch BR8	BR8-Ubuntu - 192.168.219.1	Probe source; two Internet circuits (INET1, INET2)
Branch BR5	BR5-Ubuntu - 192.168.217.5	Probe source; Internet circuit BR5-INET
Target	DC1 - 192.168.203.100	Probe responder and TCP application server
Fault injection	vyosrouter eth10 (DC1-INET-221), eth11 (DC1-MPLS-191)	Latency, then shutdowns
TCP applications	Onprem8083 (reference), app-erp-tx 8105, OnPrem8091 8443, 8085/8089/8090/8096	Application-level impact
Observation	Prisma SD-WAN Flow Browser; Stigix test and app counters	Paths, loss, latency, RTT, timeouts, reconnects

Initial conditions. vyosrouter eth0 (ALL-MPLS-190) carried a pre-existing 250 ms delay, which was left untouched. At baseline, BR8 used different circuits per traffic class: probes and voice on BR8-INET1, custom TCP on BR8-INET2, both to DC1-INET.

3. Methodology

Both probes were started three seconds apart, before any fault, so that each branch experienced the same events. Faults were applied at intervals of roughly 30 to 60 seconds, the gap depending on the time needed to collect observations from both sites. TCP impact was measured as the change in each application's cumulative timeout and reconnect counters between baseline and the end of the test, along with its round-trip time at each stage.

Step	Action	Purpose
0	Baseline of paths, probe latency and TCP counters	Nominal reference for both branches
1	Start UDP 6200 probes BR8 -> DC1 and BR5 -> DC1	Continuous measurement on both sites
2	Add 50 ms latency on DC1 Internet (eth10)	Degradation response
3	Shut DC1 MPLS (eth11)	Single transport loss
4	Shut DC1 Internet (eth10)	Loss of all direct DC1 transports
5	Stop probes, collect metrics and TCP counters	Quantify impact
6	Clear latency, restore eth11 and eth10	Return to nominal, observe failback

4. Timeline of events (UTC)

Time	Event	BR8 path to DC1	BR5 path to DC1
22:10:46 / :49	Probes started	BR8-INET1 -> DC1-INET	BR5-INET -> DC1-INET
≈22:11:25	+50 ms on DC1-INET	Unchanged	Unchanged
≈22:12:25	DC1-MPLS shut	Unchanged	Unchanged
≈22:13:07	DC1-INET shut	-	-
22:13:08.935	SD-WAN decision (BR8)	BR8-INET1 -> DC2-INET	-
by 22:13:24	BR5 on backup path	BR8-INET1 -> DC2-INET	BR5-INET -> DC2-INET
≈22:14:05	Probes stopped	-	-
≈22:14:40-22:15:00	Latency cleared, eth11 and eth10 restored	-	-
22:15:24	BR8 failback	BR8-INET2 -> DC1-INET	-
22:15:30	BR5 failback	-	BR5-INET -> DC1-INET

Times marked ≈ are command execution times; other timestamps come from the SD-WAN Flow Browser. The exact decision time for BR5 was not captured, so only the first observation on the backup path is given.

5. Results

5.1 Network probes (UDP 6200)

Metric	BR8 -> DC1	BR5 -> DC1
Baseline latency	6.3 ms	6.4 ms
Running avg. after +50 ms	28.0 ms	30.0 ms
Running avg. after MPLS shut	41.7 ms	42.8 ms
Loss before final failover	0 %	0 %
Packets sent / received	9,994 / 9,558	10,123 / 9,720
Packets lost	436 (4.4 %)	403 (4.0 %)
Equivalent outage at 50 pps	≈ 8.7 s	≈ 8.1 s
Average latency / jitter (whole test)	32.8 ms / 6.9 ms	32.4 ms / 3.7 ms

5.2 Custom TCP applications

Counters are cumulative since application start; the deltas below cover the test window only.

Application	Site	Timeouts	Reconnects	Avg RTT: before / +50 ms / after
Onprem8083	BR8	26 -> 30 (+4)	14 -> 16 (+2)	6.6 / 34.5 / 13.1 ms
app-erp-tx (8105)	BR8	26 -> 30 (+4)	14 -> 16 (+2)	14.0 / 67.5 / 14.3 ms
OnPrem8091 (8443)	BR8	24 -> 28 (+4)	14 -> 16 (+2)	10.7 / 26.3 / 15.3 ms
Onprem8085, 8089, 8090, 8096	BR8	+4 each	+2 each	400-800 ms by design
Onprem8083	BR5	41 -> 45 (+4)	22 -> 24 (+2)	15.6 / - / 15.6 ms
OnPrem8091 (8443)	BR5	40 -> 43 (+3)	21 -> 23 (+2)	- / - / 15.4 ms
Onprem8090, 8096	BR5	+4 each	+2 each	500-800 ms by design

Onprem8083 is the reference application: its sessions to DC1 were confirmed in the Flow Browser for both branches (BR8 via BR8-INET2 -> DC1-INET, BR5 via BR5-INET -> DC1-INET). The identical +2

reconnects on every active application, matching the two sessions each application keeps per peer, indicate that each application's sessions to DC1 were reset once. No application errors were recorded, and throughput resumed on new sessions without intervention.

6. Findings and analysis

F1 - Consistent behaviour across branches. BR8 and BR5 responded in the same way to every stage, and their outages at the final failover differ by only about 0.6 s. This points to a DC-side convergence mechanism rather than a branch-specific one.

F2 - Degradation tolerated without steering. +50 ms on DC1 Internet raised probe latency from about 6 ms to over 40 ms (running average) and TCP RTT by a similar margin, with no path change on either site. The SLA policy tolerates this level of degradation.

F3 - Fast control-plane decision, slower data-plane recovery. BR8's new path was chosen about two seconds after the fault, yet each probe lost about 8 seconds of traffic. As in the previous single-branch test (9.1 s), the likely contributors are tunnel liveness detection and return-path convergence from DC1 through the DCI and DC2. This should be confirmed with per-direction measurements.

F4 - TCP sessions do not survive the DC failover. Every active application reset its sessions to DC1 once and reconnected automatically. For applications with long-lived sessions (databases, ERP, file transfers), this means a visible reconnection at the application level, not just a brief pause.

F5 - Per-application path policy. On BR8, custom TCP traffic used BR8-INET2 while probes and voice used BR8-INET1. Application-based steering is active and should be kept in mind when interpreting per-flow results.

F6 - Asymmetric fallback on BR8. BR5 returned to its original path, but BR8 returned to DC1 on BR8-INET2 instead of BR8-INET1. This is harmless, but if a specific circuit is preferred for this traffic, the path policy or tunnel recovery timers should be checked.

F7 - Prior impact detected. TCP sessions from both branches to DC1 had also restarted at 18:17:23-24, during the earlier single-branch test. BR5 was therefore affected then too, which confirms that a DC1 transport outage affects every branch.

7. Recommendations

R1. Reduce the ≈ 8 s recovery: tune tunnel liveness and probe timers on the DC1 ION devices, and enable fast failure detection (for example BFD) on the routing between DC1 and DC2 across the DCI. Re-test to measure the gain.

R2. Assess the business impact of TCP session resets for critical applications, and confirm they reconnect cleanly (connection pooling, retry logic).

R3. Review the SLA thresholds for latency-sensitive applications if a +50 ms degradation should trigger steering.

R4. Verify the intended path preference on BR8 after fallback (INET1 vs INET2).

R5. Re-run the scenario with the pre-existing 250 ms MPLS delay removed to validate a nominal MPLS fallback.

8. Restoration and lab state

All changes were reverted and verified against the router's live state: eth10 and eth11 are administratively up with no QoS policy, the pre-existing eth0 delay is unchanged, and no firewall blocks were added. Both branches returned to DC1 directly.

Action	VyOS CLI equivalent	Status
Latency eth10	set traffic-policy network-emulator LAB_COMBINED_eth10 network-delay 50	Removed
Shut eth11	set interfaces ethernet eth11 disable	Reverted
Shut eth10	set interfaces ethernet eth10 disable	Reverted

Measurement limitations. The TCP session detail tool was unavailable during the test, so TCP impact was derived from application counters. Long-lived flows appear in the Flow Browser only once their records close, so paths were tracked using short-lived probe flows sharing the same tunnels.